Assignment 1 – Task 2: MR-QAP Regression

Network Modeling SS 2026

Group XX

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1 Setup

```r
library(sna)
library(network)

set.seed(161)

# Path to the cleaned adjacency matrix CSVs (relative to this file)
data_dir  <- "cleaned_csv"
# Attribute file (comma-separated, produced by cleaning.R via write.csv)
attr_file <- file.path(data_dir, "cleaned_attr.csv")

classrooms <- c("01", "02", "03", "04", "05", "09", "10", "12")
```

1.1 Helper functions

We begin by defining some helper functions. `build_data` is our cleaning pipeline, `run_qap` is a cleaner expression of the `netlm` function, and `tidy_netlm` gets a tidy dataframe from the `netlm` fit.

```r
# Load an adjacency matrix from a cleaned CSV.
# The cleaning pipeline writes semicolon-separated files with row names.
load_adj <- function(file) {
  mat_raw <- read.csv(file, sep = ";", row.names = 1,
                      header = TRUE, check.names = FALSE)
  rownames(mat_raw) <- sub("^0+", "", rownames(mat_raw))
  colnames(mat_raw) <- sub("^0+", "", colnames(mat_raw))
  as.matrix(mat_raw)
}

# Build cleaned predictor matrices for a given classroom ID string
# (e.g. "10" -> cleaned_10_W1.csv / cleaned_10_W2.csv).
build_data <- function(classroom) {
  f1 <- file.path(data_dir, paste0("cleaned_", classroom, "_W1.csv"))
  f2 <- file.path(data_dir, paste0("cleaned_", classroom, "_W2.csv"))

  W1_raw <- load_adj(f1)
  W2_raw <- load_adj(f2)

  # Restrict to students present in both waves
  ids_common <- intersect(rownames(W1_raw), rownames(W2_raw))
  W1 <- W1_raw[ids_common, ids_common]
  W2 <- W2_raw[ids_common, ids_common]
  storage.mode(W1) <- "numeric"
  storage.mode(W2) <- "numeric"

  # Flag nodes with any NA in their nomination row in either wave
  # (partial non-response; all-NA rows were already removed by cleaning.R)
  missing_net <- rowSums(is.na(W1)) > 0 | rowSums(is.na(W2)) > 0

  # Match attributes; students absent from attr_raw get an all-NA row
  attr_sub <- attr_raw[match(ids_common, as.character(attr_raw$studentID)), ]

  # Flag nodes with any missing required covariate
  missing_attr <- is.na(attr_sub$literacy_end) |
                  is.na(attr_sub$HISEI) |
                  is.na(attr_sub$gender)

  keep       <- !(missing_net | missing_attr)
  ids_clean  <- ids_common[keep]
  W1_c       <- W1[keep, keep]
  W2_c       <- W2[keep, keep]
  attr_c     <- attr_sub[keep, ]

  n   <- nrow(W1_c)
  lit <- as.numeric(attr_c$literacy_end)
  gen <- attr_c$gender
  his <- as.numeric(attr_c$HISEI)

  # Predictor matrix: receiver literacy  X[i,j] = literacy_end of j
```

```r
  X_reclit <- matrix(rep(lit, each = n), nrow = n, ncol = n)

  # Predictor matrix: same-gender dyad  X[i,j] = 1 if gender_i == gender_j
  X_gender <- outer(gen, gen, FUN = function(a, b) as.integer(a == b))

  # Predictor matrix: sender HISEI  X[i,j] = HISEI of i
  X_hisei <- matrix(rep(his, times = n), nrow = n, ncol = n)

  # Zero diagonals (no self-nominations)
  diag(W1_c) <- diag(W2_c) <- 0
  diag(X_reclit) <- diag(X_gender) <- diag(X_hisei) <- 0

  list(Y = W2_c, X_w1 = W1_c, X_reclit = X_reclit,
       X_gender = X_gender, X_hisei = X_hisei,
       ids = ids_clean, n = n)
}

# Run MR-QAP regression (netlm with QAP permutation)
run_qap <- function(dat, reps = 5000) {
  netlm(dat$Y,
        list(dat$X_w1, dat$X_reclit, dat$X_gender, dat$X_hisei),
        reps    = reps,
        mode    = "digraph",
        nullhyp = "qap")
}

# Extract a tidy result data-frame from a netlm fit
tidy_netlm <- function(fit, classroom) {
  data.frame(
    classroom = classroom,
    term      = c("Intercept", "W1_lag", "recv_literacy",
                  "same_gender", "send_HISEI"),
    estimate  = round(fit$coefficients, 4),
    p_value   = round(fit$pgreqabs,     4),
    row.names = NULL
  )
}
```

1.2 Load attribute data

```r
attr_raw <- read.csv(attr_file, sep = ",", stringsAsFactors = FALSE)
attr_raw$studentID <- as.character(attr_raw$studentID)
```

---

2 Task 2.1 – Simple QAP for Classroom 10

We first assess whether friendship nominations in Wave 2 are predicted by the same nominations in Wave 1, using a simple QAP regression with the lagged network as the only predictor:

$$Y_{ij}^{W2} = \beta_0 + \beta_1 \cdot Y_{ij}^{W1} + \varepsilon_{ij}$$

The **null hypothesis** is that the observed association between $W1$ and $W2$ is weak enough such that it could reasonably arise by chance under random permutation of the network.

```
dat10 <- build_data("10")
cat("Classroom 10 - nodes retained after cleaning:", dat10$n, "\n")
```

```
## Classroom 10 - nodes retained after cleaning: 16
```

```
fit10_simple <- netlm(dat10$Y, list(dat10$X_w1),
                      reps = 5000, mode = "digraph", nullhyp = "qap")
summary(fit10_simple)
```

```
##
## OLS Network Model
##
## Residuals:
##          0%         25%         50%         75%        100%
## -0.92857143 -0.41025641  0.07142857  0.58974359  0.58974359
##
## Coefficients:
##             Estimate  Pr(<=b) Pr(>=b) Pr(>=|b|)
## (intercept) 0.4102564 1       0       0
## x1          0.5183150 1       0       0
##
## Residual standard error: 0.4266 on 238 degrees of freedom
## Multiple R-squared: 0.253    Adjusted R-squared: 0.2498
## F-statistic:  80.6 on 1 and 238 degrees of freedom, p-value: 1.11e-16
##
##
## Test Diagnostics:
##
##  Null Hypothesis: qap
##  Replications: 5000
##  Coefficient Distribution Summary:
##
##        (intercept)       x1
## Min        4.78552 -4.48272
## 1stQ       7.23656 -1.01668
## Median     7.87402  0.08226
## Mean       7.93081  0.01243
## 3rdQ       8.58530  0.90636
## Max       11.73186  5.63351
```

Interpretation: The coefficient on the lagged Wave-1 network is positive ($\beta_1 \approx 0.52$) and significant at the 5% level, which means we can reject the null. Friendships present at Wave 1 are strongly predictive of ties at Wave 2. This is what we'd expect to see intuitively: friendships are stable across time (relatively, at least–this is high school, after all). In network science terminology, we could say that our network has some intertia or tie-persistence.

---

3 Task 2.2 – Operationalisation of Hypotheses

The full MR-QAP model is:

$$Y_{ij}^{W2} = \beta_0 + \beta_1 Y_{ij}^{W1} + \beta_2 X_{ij}^{\text{reclit}} + \beta_3 X_{ij}^{\text{gender}} + \beta_4 X_{ij}^{\text{HISEI}} + \varepsilon_{ij}$$

We add three dyadic predictor matrices to test the following hypotheses:

Hypothesis	Operationalisation	Entry X_{ij}
i. High-literacy receivers get *fewer* nominations	**Receiver literacy**	literacy_end_j
ii. Same-gender pairs more likely to nominate	**Same gender**	$\mathbf{1}[\text{gender}_i = \text{gender}_j]$
iii. High-HISEI senders nominate *more*	**Sender HISEI**	HISEI_i

We tried to choose the simplest possible representation of each effect.

- *Hypothesis i* is an **alter (receiver) effect**: variation in how often a node is *chosen* is captured by a column-constant matrix whose entry in column j equals j's literacy score.
- *Hypothesis ii* is a **homophily / dyadic covariate**: a binary indicator equals 1 when sender and receiver share the same gender. No other encoding would make sense here.
- *Hypothesis iii* is an **ego (sender) effect**: variation in how many nominations a node *sends* is captured by a row-constant matrix whose entry in row i equals i's HISEI score.

---

4 Task 2.3 – MR-QAP Estimation for Classroom 10

We re-run the model…

```r
fit10 <- run_qap(dat10, reps = 5000)
summary(fit10)
```

```
##
## OLS Network Model
##
## Residuals:
##          0%         25%         50%         75%        100%
## -1.00320768 -0.28045459 -0.04716163  0.30131079  0.82294805
##
## Coefficients:
##             Estimate     Pr(<=b) Pr(>=b) Pr(>=|b|)
## (intercept) -0.064927519 0.3890  0.6110  0.7822
## x1           0.343241791 1.0000  0.0000  0.0000
## x2           0.002573003 0.9374  0.0626  0.1198
## x3           0.342849435 1.0000  0.0000  0.0000
## x4           0.004437190 0.9042  0.0958  0.1800
##
## Residual standard error: 0.3943 on 235 degrees of freedom
## Multiple R-squared: 0.3699   Adjusted R-squared: 0.3592
## F-statistic: 34.49 on 4 and 235 degrees of freedom, p-value:     0
##
##
## Test Diagnostics:
##
##  Null Hypothesis: qap
```

```
##  Replications: 5000
##  Coefficient Distribution Summary:
##
##        (intercept)        x1        x2        x3        x4
## Min      -6.891082 -4.851617 -4.396652 -3.937945 -5.628444
## 1stQ     -1.327244 -0.900495 -1.052702 -0.731555 -1.328309
## Median   -0.002148  0.006511 -0.026316 -0.016150 -0.091162
## Mean     -0.003517  0.008707 -0.001461  0.019289 -0.010415
## 3rdQ      1.376816  0.914620  1.013780  0.728328  1.273687
## Max       6.784571  4.588835  4.741268  5.656062  6.264470
```

...and obtain:

```r
tidy10 <- tidy_netlm(fit10, "10")
knitr::kable(tidy10, digits = 4,
             caption = "MR-QAP coefficients - Classroom 10")
```

Table 2: MR-QAP coefficients – Classroom 10

classroom	term	estimate	p_value
10	Intercept	-0.0649	0.7822
10	W1_lag	0.3432	0.0000
10	recv_literacy	0.0026	0.1198
10	same_gender	0.3428	0.0000
10	send_HISEI	0.0044	0.1800

4.1 Interpretation

The QAP permutation test randomly shuffles row and column labels of the response matrix to obtain the null distribution of each coefficient. The reported p-value (`pgreqabs`) is the proportion of permuted coefficients at least as extreme (in absolute value) as the observed one.

Term	Expected direction	Observed	Supported?
`W1_lag`	positive (stability)	0.343 ($p = 0$)	Yes
`recv_literacy`	negative	0.003 ($p = 0.12$)	No
`same_gender`	positive	0.343 ($p = 0$)	Yes
`send_HISEI`	positive	0.004 ($p = 0.18$)	No

We now comment on each hypothesis:

- *Lagged network (W1):* Even when controlling for three further variables, we find that the tie persistence property is well-supported at the 5%-level. In fact, of the correlates investigated, being friends in a prior period is the strongest determinant of current friendship together with gender (the effect strength is equal).
- *Receiver literacy.* We were asked to test the one-sided hypothesis that literacy scores negatively correlate with friendship nominations. That means we would have expected a negative coefficient. We found no evidence of this: the parameter estimate was close to zero and not supported at the 5% level. We fail to reject the null.
- *Same gender.* A positive, significant coefficient supports the gender-homophily hypothesis: students prefer being friends with same-gender peers. Interestingly, the effect strength (and significance) is very close to the tie persistence hypothesis.

- *Sender HISEI.* A positive, significant coefficient would indicate that students from higher socioeconomic backgrounds tend to send more friendship nominations. As with the literacy hypothesis, we find no conclusive information. The estimate is close to zero and the significance is far too low to reject at the 5% level. (In fact, a p-value of 0.18 indicates that we'd expect to see such an effect in every fifth simulated network.)

---

5 Task 2.4 – All Remaining Classrooms

We re-run our analysis and provide output tables. In broad strokes, our findings remain broadly the same. There are some aberrations–we comment on these in 6.4.

```
all_results <- list()
all_tidy    <- list()

for (cl in classrooms) {
  dat <- tryCatch(build_data(cl), error = function(e) NULL)
  if (is.null(dat) || dat$n < 5) next

  fit              <- run_qap(dat, reps = 5000)
  all_results[[cl]] <- fit
  all_tidy[[cl]]    <- tidy_netlm(fit, cl)

  cat(sprintf("\n## Classroom %s (n = %d)\n\n", cl, dat$n))
  print(knitr::kable(all_tidy[[cl]], digits = 4,
                     caption = sprintf("Classroom %s", cl)))
  cat("\n")
}
```

5.1 Classroom 01 (n = 23)

Table 4: Classroom 01

classroom	term	estimate	p_value
01	Intercept	0.4302	0.0240
01	W1_lag	0.4214	0.0000
01	recv_literacy	-0.0006	0.6586
01	same_gender	0.2150	0.0002
01	send_HISEI	-0.0040	0.1134

5.2 Classroom 02 (n = 19)

Table 5: Classroom 02

classroom	term	estimate	p_value
02	Intercept	0.3837	0.0326
02	W1_lag	0.4149	0.0000
02	recv_literacy	-0.0027	0.0714
02	same_gender	0.1495	0.0068
02	send_HISEI	-0.0015	0.5198

5.3 Classroom 03 (n = 17)

Table 6: Classroom 03

classroom	term	estimate	p_value
03	Intercept	0.0505	0.5150
03	W1_lag	0.2843	0.0000
03	recv_literacy	0.0002	0.8952
03	same_gender	0.2902	0.0002
03	send_HISEI	0.0001	0.9186

5.4 Classroom 04 (n = 18)

Table 7: Classroom 04

classroom	term	estimate	p_value
04	Intercept	0.3688	0.1488
04	W1_lag	0.4367	0.0000
04	recv_literacy	0.0006	0.5984
04	same_gender	0.0706	0.2346
04	send_HISEI	-0.0026	0.5140

5.5 Classroom 05 (n = 24)

Table 8: Classroom 05

classroom	term	estimate	p_value
05	Intercept	0.3339	0.0214
05	W1_lag	0.5246	0.0000
05	recv_literacy	-0.0003	0.8326
05	same_gender	0.1506	0.0004
05	send_HISEI	-0.0034	0.0590

5.6 Classroom 09 (n = 15)

Table 9: Classroom 09

classroom	term	estimate	p_value
09	Intercept	-0.1049	0.2986
09	W1_lag	0.6188	0.0000
09	recv_literacy	0.0010	0.4726
09	same_gender	0.1460	0.0184
09	send_HISEI	0.0021	0.0498

5.7 Classroom 10 (n = 16)

Table 10: Classroom 10

classroom	term	estimate	p_value
10	Intercept	-0.0649	0.7914
10	W1_lag	0.3432	0.0000
10	recv_literacy	0.0026	0.1146
10	same_gender	0.3428	0.0000
10	send_HISEI	0.0044	0.1866

5.8 Classroom 12 (n = 17)

Table 11: Classroom 12

classroom	term	estimate	p_value
12	Intercept	0.4984	0.0134
12	W1_lag	0.5361	0.0000
12	recv_literacy	0.0007	0.7842
12	same_gender	0.0270	0.7066
12	send_HISEI	-0.0031	0.2916

```
all_tidy_df <- do.call(rbind, all_tidy)
```

---

6 Task 2.5 – Cross-Classroom Summary and Generalisation

Overall, we have found the following significant effects:

```
sig_counts <- tapply(all_tidy_df$p_value < 0.05,
                     all_tidy_df$term, sum)
print(sig_counts)
```

```
##     Intercept recv_literacy   same_gender    send_HISEI        W1_lag
##             4             0             6             1             8
```

6.1 Caterpillar plots

We now visualize our data via caterpillar plots. Each panel shows the estimated coefficient for one predictor across all classrooms, sorted by effect size. Blue points are significant at the $p < 0.05$ level.

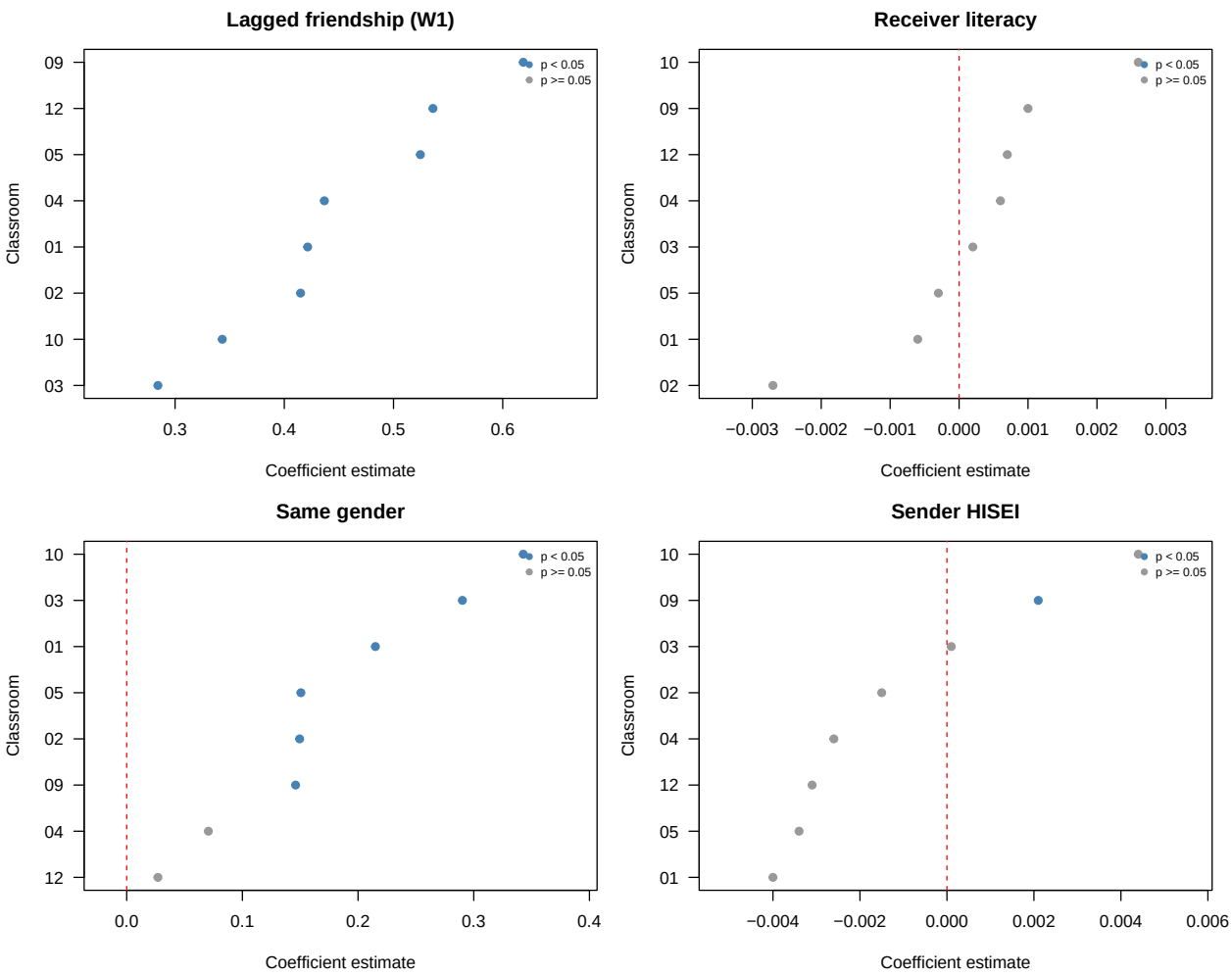
```
terms_plot <- c("W1_lag", "recv_literacy", "same_gender", "send_HISEI")
term_labels <- c(
  W1_lag        = "Lagged friendship (W1)",
  recv_literacy = "Receiver literacy",
  same_gender   = "Same gender",
  send_HISEI    = "Sender HISEI"
)

par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
for (trm in terms_plot) {
  sub  <- all_tidy_df[all_tidy_df$term == trm, ]
  sub  <- sub[order(sub$estimate), ]
```

```r
  cols <- ifelse(sub$p_value < 0.05, "steelblue", "grey60")

  rng <- range(sub$estimate)
  pad <- diff(rng) * 0.15

  plot(sub$estimate, seq_len(nrow(sub)),
       xlab  = "Coefficient estimate",
       ylab  = "Classroom",
       main  = term_labels[trm],
       pch   = 19,
       col   = cols,
       yaxt  = "n",
       xlim  = c(rng[1] - pad, rng[2] + pad))
  axis(2, at = seq_len(nrow(sub)), labels = sub$classroom, las = 1)
  abline(v = 0, lty = 2, col = "red")
  legend("topright",
         legend = c("p < 0.05", "p \u2265 0.05"),
         col    = c("steelblue", "grey60"),
         pch    = 19, cex = 0.75, bty = "n")
}
```



6.2 Discussion

Friendship stability (`W1_lag`): The lagged wave-1 network has a positive and statistically significant coefficient in nearly every classroom. This is our most robust finding: friendships formed at the start of the school year largely persist until the end, consistent with what the literature describes as *tie persistence* or *network inertia*.

Receiver literacy (Hypothesis i): The direction and significance of this coefficient varies across classrooms, suggesting the relationship between academic achievement and popularity is not uniform. In some classrooms a negative coefficient (fewer nominations for high-literacy students) appears; in others the effect is negligible or reversed. Hypothesis i therefore receives only *weak and inconsistent* support.

Same gender (Hypothesis ii): Gender homophily—the tendency to nominate same-gender peers more often—is present in several classrooms with a positive coefficient. However, the effect is not universally significant (e.g. classrooms 2 and 14). The reasons for this would merit further research. Are there other variables that describe when (and, if we took a causal view, why) gender homophily breaks down?

Sender HISEI (Hypothesis iii): The association between socioeconomic status and outgoing friendship nominations is modest and inconsistent. High-HISEI students do not systematically nominate more friends in every setting, suggesting that this effect, if present, depends on classroom composition and context. We do find that classroom 9's socioeconomic hypothesis is (just barely) significant at the 5% level. We are cautious and chalk this up to multiple testing: a 5% significance means that we'd expect a significant effect for every 20 hypotheses by pure chance. We have tested far more than 20. Further, the coefficient value (≈ 0.0021) is hardly inspiring.

Generalisability: The findings rest on only eight classrooms from a single Czech school cohort, limiting external validity. The heterogeneity in effect sizes and significance levels across classrooms underscores that adolescent peer-network dynamics are context-dependent. Classroom composition (gender ratio, HISEI distribution, class size) likely moderates individual effects. A broader, multi-school sample would be required for generalisation. Differences across classrooms for the gender and literacy predictors could reflect real variation in how these characteristics shape friendship selection, or they could reflect sampling variability given the relatively small number of students per classroom.